Quadratics & Polynomials

ANSWER KEY

Section 1: Factoring & Quadratic Formula

1 What is the positive solution to the quadratic equation $3x^2 - 10x - 8 = 0$?

a. 2

b. 4 ✓

c. $\frac{2}{3}$

d. $\frac{4}{3}$

Factor by grouping: $(3x + 2)(x - 4) = 0$. The roots are $x = -\frac{2}{3}$ and $x = 4$. The positive solution is 4.

2 If $x^2 - 6x + k = 0$ has two equal real roots, what is the value of k?

a. 3

b. 6

c. 9 ✓

d. 36

Equal real roots require discriminant $D = b^2 - 4ac = 0$: $(-6)^2 - 4(1)(k) = 0 \Rightarrow 36 - 4k = 0 \Rightarrow k = 9$.

3 What is the sum of the solutions of the quadratic equation $2x^2 + 8x - 15 = 0$?

a. -4 ✓

b. 4

c. -7.5

d. 8

By Vieta's formula, sum of roots is $-\frac{b}{a} = -\frac{8}{2} = -4$.

Section 2: Vertex Form & Parabola Extremes

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4 What is the minimum value of the quadratic function $f(x) = 2x^2 - 12x + 23$?

a. 3

b. 5 ✓

c. 7

d. 11

Vertex x-coordinate $= -\frac{-12}{2(2)} = 3$. Minimum value $= f(3) = 2(9) - 12(3) + 23 = 18 - 36 + 23 = 5$.

5 The parabola $y = -(x - 4)^2 + 16$ models the flight path of a projectile. What is the maximum height achieved?

a. 4

b. 8

c. 16 ✓

d. 32

In vertex form $y = a(x - h)^2 + k$, since $a = -1 < 0$, the maximum occurs at vertex $(4, 16)$, giving maximum height 16.

6 What is the axis of symmetry of the parabola $y = x^2 + 8x - 5$?

a. $x = -4$ ✓

b. $x = 4$

c. $x = -8$

d. $y = -4$

Axis of symmetry is $x = -\frac{b}{2a} = -\frac{8}{2(1)} = -4$.

Section 3: Polynomial Division & Factors

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7 What is the remainder when $P(x) = x^3 - 4x^2 + 5x - 7$ is divided by $(x - 2)$?

a. -5 ✓

b. -3

c. 3

d. 5

By Remainder Theorem, evaluate $P(2) = 2^3 - 4(2^2) + 5(2) - 7 = 8 - 16 + 10 - 7 = -5$.

8 If $(x - 3)$ is a factor of $x^3 - 2x^2 - kx + 12$, what is the value of k?

a. 3

b. 4

c. 7 ✓

d. 9

Factor means $P(3) = 0$:
$3^3 - 2(3^2) - 3k + 12 = 0 \Rightarrow 27 - 18 - 3k + 12 = 0 \Rightarrow 21 - 3k = 0 \Rightarrow 3k = 21 \Rightarrow k = 7$.

9 Which of the following polynomials has roots at $x = -1, x = 2,$ and $x = 3$?

a. $x^3 - 4x^2 + x + 6$ ✓

b. $x^3 - 4x^2 - x + 6$

c. $x^3 + 4x^2 + x - 6$

d. $x^3 - 2x^2 - 5x + 6$

Expand $(x + 1)(x - 2)(x - 3) = (x^2 - x - 2)(x - 3) = x^3 - 3x^2 - x^2 + 3x - 2x + 6 = x^3 - 4x^2 + x + 6$
.

Section 4: Advanced Polynomial Graphs & Vieta

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10 What are all real roots of the polynomial equation $x^4 - 5x^2 + 4 = 0$?

a. $x = \pm 1, \pm 2$ ✓

b. $x = \pm 1, \pm 4$

c. $x = 1, 2$

d. $x = 0, 1, 4$

Factor as quadratic in x^2: $(x^2 - 1)(x^2 - 4) = 0 \Rightarrow x^2 = 1 \Rightarrow x = \pm 1$, and $x^2 = 4 \Rightarrow x = \pm 2$.

11 What is the product of the roots of the cubic equation $2x^3 - 6x^2 + 4x - 10 = 0$?

a. 2

b. 3

c. 5 ✓

d. -5

By Vieta's formulas, product of roots of $ax^3 + bx^2 + cx + d = 0$ is $-\frac{d}{a} = -\frac{-10}{2} = 5$.

12 How many times does the graph of $f(x) = (x - 1)^2(x + 3)$ cross the x-axis?

a. 1 **time (crosses at** $x = -3$**, touches at** $x = 1$**)** ✓

b. 2 times

c. 3 times

d. 0 times

Root $x = 1$ has even multiplicity 2 (tangent/bounces without crossing). Root $x = -3$ has odd multiplicity 1 (crosses). Thus it crosses the x-axis exactly 1 time.

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